Two Axioms Remain:
An Invitation to the Next Generation

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For the student who hasn't started yet.

What This Is

We built something called the Cathedral: a 78-file computer program that proves, with mathematical certainty, that the Riemann Hypothesis is *exactly equivalent* to the convergence of a specific approximation problem. Every step is verified by a computer. Every assumption is named. Every failed attempt is preserved.

The proof is complete—except for two assumptions, called *axioms*. We couldn't prove them. We named them, classified them, and documented exactly what they say and why they're hard.

This paper is an invitation. Those two axioms are now your problem.

Who Built This

A computer scientist in New Mexico. Not a professional mathematician. No PhD in mathematics. No university affiliation. No research grant.

The tools: a laptop, an internet connection, Lean 4 (free, open source), Mathlib (free, open source), and AI assistance (commercially available).

The timeline: 23 days.

If you have a laptop and curiosity, you have everything that was needed to build this. The barrier to entry for machine-verified mathematics is now approximately zero.

What You Need to Know

The Problem

The Riemann Hypothesis (1859) says: all non-trivial zeros of the Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ lie on the line $\mathrm{Re}(s) = \frac{1}{2}$.

Nobody has proved this. Nobody has disproved it. It has been open for 167 years. There is a \$1,000,000 prize.

The Translation

The Cathedral proves that this is equivalent to:

$$\forall\,\varepsilon > 0,\ \exists\,N_0,\ \forall\,N \geq N_0,\ \exists\,v \in \mathbb{R}^{N-1}: \quad \int_0^1 \left(1 - \sum_{k=2}^{N} v_k \left\{\frac{1}{kx}\right\}\right)^2 dx < \varepsilon.$$

In words: you can approximate the constant function 1 arbitrarily well by mixing together sawtooth waves $\{1/(kx)\}$.

The Two Axioms

1. `bd_mellin_at_zero`: The Mellin transform of the fractional-part function $\{1/(kx)\}$, which is proved for $\mathrm{Re}(s) > 1$ by direct computation, extends by analytic continuation to $\mathrm{Re}(s) > 0$.

   What this means: a specific integral formula, proved for large values of a parameter, also holds for smaller values. This is a standard result in complex analysis, but formalizing it in Lean requires Mathlib's analytic continuation infrastructure.

2. `rh_implies_l2_convergence`: If the Riemann Hypothesis is true, then the Báez-Duarte distance $d_N^2 \to 0$.

   What this means: assuming RH, you can construct explicit weights v_k that make the integral above as small as you want. The construction uses the Mertens function, Abel summation, and the Montgomery–Vaughan mean value theorem.

Axiom 1 is a formalization challenge: the mathematics is understood, but translating it into Lean 4 requires careful work with analytic continuation.

Axiom 2 is a *mathematics* challenge: it packages several deep results in analytic number theory. Proving it formally would require formalizing the Mertens–Abel–Montgomery–Vaughan chain.

What You Can Do Right Now

If You Know Some Lean 4

1. **Prove Axiom 1.** This is the more accessible target. You need Mathlib's `DifferentiableOn.analyticCo` or equivalent, applied to the Mellin integral $\int_0^1 \{1/(kx)\}\, x^{s-1}\, dx$. The integral is absolutely convergent for $\mathrm{Re}(s) > 0$; the challenge is showing that the closed-form expression (involving $\zeta(s)$ and $1/(s-1)$) extends analytically.

2. **Prove one of the 38 non-critical axioms.** The Cathedral contains 40 axioms total. Only 2 are on the crown path. The other 38 support alternative proof engines (spectral, sieve, Vasyunin). Proving any one of them strengthens the overall formalization.

3. **Improve an existing proof.** Some of the Cathedral's 644 theorems use `omega` or `norm_num` where a more elegant proof exists. Professional Lean hackers will see opportunities to improve proof clarity and performance.

If You Know Some Analysis

1. **Decompose Axiom 2.** The current axiom is a "black box" that packages the entire forward direction. A useful contribution would be to decompose it into 3–5 intermediate lemmas, each of which is independently meaningful and potentially formalizable.

2. **Find a better forward proof.** The Báez-Duarte theorem uses the Mertens function bound $|M(x)| = O(x^{1/2} \log^2 x)$ under RH. Is there a more direct path from RH to $d_N^2 \to 0$ that avoids the Mertens detour?

3. **Study the decay rate.** The Cathedral's numerical experiments show $d_N^2 \sim c/\ln N$ with $c \approx 1/21.65$. Can you prove this asymptotic? Can you prove a weaker bound, like $d_N^2 = o(1)$, from RH with fewer tools?

If You Know Some Physics

1. **Identify the operator.** The Gram matrix G_N is the finite-dimensional shadow of some infinite-dimensional operator whose spectral properties encode RH. What is this operator? The eigenvalue data ($\lambda_{\min} \sim C/\ln N$, $\lambda_{\max} \sim N$, GUE-like statistics) constrain it.

2. **Interpret the constant.** The quantum stiffness $C \approx 21.65 = 1/(2 + \gamma - \ln 4\pi)$ appears to be a physical constant of the "prime number vacuum." Does it arise naturally from any known quantum system?

What Modern Math Research Actually Looks Like

It doesn't look like a genius working alone in a garret.

It looks like this:

1. **Day 1–7**: Explore. Try things. Fail. Archive the failures. We produced 96 files of failed attempts. Every failure taught us something.

2. **Day 8–14**: Find the right framework. For us, this was the Parseval Bridge—the realization that frequency- domain methods bypass the hardest discrete computations. This came from a failure: the False Dedekind Reciprocity at $(a, b) = (3, 2)$.

3. **Day 15–21**: Build. Once the framework is right, construction is fast. AI handles the mechanical code generation. The human provides architecture. The compiler verifies everything.

4. **Day 22–23**: Audit. Check every file. Fix every docstring. Remove every ghost axiom. Reduce, simplify, clean. The Great Audit reduced 178 files to 78 active + 96 archived.

The ratio of exploration to construction is roughly 2:1. Two weeks of wandering, one week of building. This is normal. If you're not failing most of the time, you're not exploring hard enough.

Why You Should Care

The Riemann Hypothesis is not just a mathematical curiosity. It controls the distribution of prime numbers, which are the atoms of arithmetic. Every encryption algorithm, every hash function, every digital signature depends on properties of primes.

But more than that: the RH is a test case for whether humans can understand the deepest structures in mathematics. If we can prove it, we learn something profound about the nature of number. If we can't, we learn something profound about the limits of mathematical reasoning.

Either way, the attempt is worth making.

A Note on Credentials

You do not need a PhD to contribute to this project. You do not need a university appointment. You do not need a research grant.

You need:

- A laptop.
- Lean 4 (free).
- Mathlib (free).
- Curiosity.
- The willingness to fail 96 times before you succeed once.

The compiler does not check your credentials. It checks your proof.

The Torch

We built the Cathedral. We mapped the territory. We named the axioms. We preserved the failures. We verified every step.

We could not prove the last two things.

Now it's your turn.

The primes have been singing for 167 years.
The Cathedral stands, waiting for its final stones.
We built it as far as we could.
The rest belongs to you.

References

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